The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science

Andre Milzarek · Junfeng Wu · Junchi Yang · Spring Semester 2025

MAT 3007 — Optimization

Exercise Sheet 3

Problem 1 (LPs and the Simplex Method): (approx. *16* pts)
We consider an LP in its standard form:

$$\min_{\boldsymbol{x}} \ \boldsymbol{c}^\top \boldsymbol{x} \qquad \text{s.t.} \qquad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}, \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$ are given. Decide whether the following statements are true or false and explain your answer (i.e., give a short proof or a counterexample).

a) We apply the two-phase simplex method to solve problem (1). Let $(\boldsymbol{x}^*, \boldsymbol{y}^*)$ be a solution of the auxiliary problem generated in phase I. Then, $(\boldsymbol{x}^*, \boldsymbol{y}^*)$ can have more than m non-zero entries.

b) Phase I of the two-phase simplex method can indicate unboundedness of problem (1).

c) Let $\boldsymbol{x}^*$ be an optimal solution of (1) and suppose that $\boldsymbol{x}^*$ has strictly less than m positive components. Then, there exists another optimal solution that is a basic feasible solution and that has m positive entries.

d) Basic solutions of (1), that are not extreme points of the set $\{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$, need to be infeasible.

Problem 2 (Application of the Simplex Method): (approx. *20* pts)
Consider the following linear program:

$$\begin{aligned} \text{maximize} \quad & x_1 + 2x_2 + 3x_3 + 4x_4 & \\ \text{subject to} \quad & x_1 + x_2 + x_3 & \leq 1 \\ & x_3 - x_4 & \leq 2 \\ & x_2 - 2x_3 + 4x_4 & \leq 3 \\ & x_1, x_2, x_3, x_4 & \geq 0. \end{aligned}$$

Apply the simplex method (simplex tableau) to solve this problem. For each step, clearly mark what is the current basis, the current basic solution, and the corresponding objective function value.

Problem 3 (Two-Phase Simplex Method: I): (approx. *22* pts)
Apply the two-phase simplex method to solve the following linear program:

$$\begin{aligned} \text{minimize} \quad & x_1 + x_2 + 2x_4 & \\ \text{subject to} \quad & x_1 - x_2 & \geq 1 \\ & x_1 + x_2 - x_3 - x_4 & \leq 2 \\ & x_2, x_3 & \geq 0. \end{aligned}$$

Problem 4 (Two-Phase Simplex Method: II): (approx. *22* pts)
Use the two-phase simplex method to completely solve the linear optimization problem:

$$\begin{aligned} \text{minimize} \quad & x_2 - 4x_3 + x_5 \\ \text{subject to} \quad & x_1 + x_2 + x_3 + x_4 = 1 \\ & x_2 + x_3 + x_4 - x_5 = -1 \\ & x_1 + 2x_3 - x_5 \geq 0 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$

Problem 5 (Simplex Tableau): (approx. *20* pts)
While solving a linear program in standard form, we arrive at the simplex tableau given in Table 1 with basic variables x_1, x_4, x_5. The entries α, β, γ, δ and η in the tableau are unknown parameters. For each of the following statements, find suitable conditions for the parameter values that will make the statement true (sufficient conditions are enough).

B	0	1	α	0	0	β	-1	0
4	0	1	0	1	0	1	0	η
5	0	-1	-1	0	1	0	δ	5
1	1	0	γ	0	0	-1	0	0

Table 1: Simplex tableau with unknown parameter

a) The current solution is feasible but not optimal.

b) The LP is unbounded (i.e., the optimal value is $-\infty$).

c) The objective function at the new basic feasible solution after one simplex iteration is -1.

d) The simplex method stops in the next iteration.

Sheet 3 is due on **Mar, 9th**. Submit your solutions before **Mar, 9th**, **11:00 pm**.